

Coding & Solution

Date	29 September 2026
Team ID	SWTID-2026-5955
Project Name	PocketSmart AI - Your Smart Budget & Recommendation Assistant
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is modular, readable, and well-commented with appropriate function and variable names.
2. The solution must demonstrate end-to-end working functionality matching the defined features of the project.
3. Follow clean code practices including error handling, validation, and security measures where applicable.
4. Include clear setup and execution instructions to reproduce and test the solution as described.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/SWTID-2026-5955/PocketSmart-AI (IBM Cloud deployment)
Programming Language(s)	Python 3.11, JavaScript, HTML5, CSS3
Framework(s) Used	Flask, FastAPI, Uvicorn, Jinja2, google-generativeai (Gemini 1.5 Flash Pro), Pillow, Flask-Session, CORS
Key Features Implemented	• User Registration/Login with JWT bccrpp /token /session-info • Home Planner /generate-home - Amazon IKEA Flipkart budget recommendations • Party Planner /generate-party - Swiggy Zomato OYO proportional allo • Jewelry Planner /generate-jewelry - outfit image upload matching • History page /history • Responsive UI, IBM Cloud hosting
Pending / Incomplete Features	• Real-time payment gateway integration • Live scraping - currently using mock APIs + Gemini • Mobile app (Flutter) - future scope

Setup / Run Instructions

1. Clone repo
2. pip install -r requirements.txt
3. Set GOOGLE_API_KEY in .env
4. python app.py or uvicorn main:app
5. Open <http://localhost:5000>

Code Quality Checklist

S.No	Criteria	Status (Yes/No)
1	Code is modular and organized into functions/classes	✓ Yes
2	Meaningful variable and function names are used	✓ Yes
3	Code includes commented/documentation where necessary	✓ Yes
4	Error handling is implemented for critical operations	✓ Yes
5	The application runs without critical errors	✓ Yes
6	Code is committed to a version control repository	✓ Yes

Additional Notes / Comments

PocketSmart AI codebase follows PEP8, modular architecture with routes/, services/, models/, gemini_uutls.py reusable across Home/Party/Jewelry. Auth module uses JWT + bcrypt for security. Gemini 1.5 Flash Pro prompts optimized for budget allocation. Fallback recommendations implemented for API failures. Code tested end-to-end with real budgets (5k-1L) across all planners. GitHub repo includes README, requirements.txt, .env example, IBM Cloud deployment instructions. Future scope: Flutter mobile app, real-time payment integration, live e-commerce scraping. Overall working solution meets all functional requirements.